Skills Summary

Rounding and Ordering on a Number Line

Use this reference while completing your worksheet or when studying.

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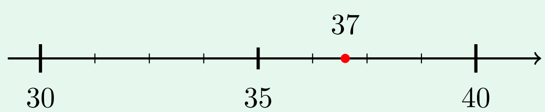
1. Rounding to the nearest ten

Draw the picture, then decide.

- Find the ten just below your number and the ten just above it. Those are the two ends of your number line.
- The middle mark is halfway between them — it always ends in 5.
- Put your number on the line. **The end it is closer to is the answer.**
- Landed exactly on the middle? The rule is: go to the bigger end.

Example: Round 37 to the nearest ten.
Step 1. Rounding asks which benchmark is nearer, so first draw the two tens that 37 sits between.
Step 2. The ends are 30 and 40, and the middle is 35. Since 37 is past 35, the right end is nearer.



⇒ **40**

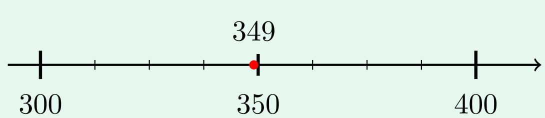
Try it: Round 62 to the nearest ten.

check: **60**

2. Rounding to the nearest hundred

Same picture, bigger hops. The ends are the hundred below and the hundred above; the middle mark ends in 50. Look only at where the number sits between those two ends — the ones digit never decides a hundreds rounding.

Example: Round 349 to the nearest hundred.
Step 1. Nearest hundred, so the benchmarks are hundreds: find the two the number sits between.
Step 2. The ends are 300 and 400 and the middle is 350. 349 is just short of the middle, so the left end is nearer.



⇒ **300**

Try it: Round 571 to the nearest hundred.

check: **600**

3. Comparing two numbers

Further right means greater. To find out which is further right without drawing every tick: compare the *hundreds* first, then the *tens*, then the *ones*. Stop at the first place where the digits differ — that place decides it, no matter what comes after.

The small point of $<$ and $>$ always aims at the smaller number.

Example: Write $<$ or $>$ between 428 and 482.
Step 1. Comparing starts at the biggest place value, so check the hundreds before anything else.
Step 2. Both have 4 hundreds, so move to the tens: 2 tens against 8 tens. So 428 sits to the left.
$\Rightarrow$ $\mathbf{428 < 482}$

Try it: Write $<$ or $>$ between 615 and 609.

check: $\mathbf{615 > 609}$

4. Putting numbers in order

Ordering is comparing, repeated. Sort into groups by the hundreds digit first, then order inside each group by the tens, then by the ones. Write the finished list left to right, the way it would read on a number line.

Careful: a longer-looking number is not automatically bigger, and the last digit never decides anything on its own.

Example: Order 204, 240 and 199 from least to greatest.
Step 1. Ordering means placing them left to right on one number line, so compare place by place, starting with the hundreds.
Step 2. 199 has 1 hundred, so it is smallest. 204 and 240 both have 2 hundreds, so compare tens: 0 against 4, putting 204 first.
$\Rightarrow$ **199, 204, 240**

Try it: Order 530, 305 and 503 from least to greatest.

check: **305, 503, 530**